How the Overview numbers are calculated

A plain-language guide to every number and chart on the Truck turnaround Overview page.

1. The basics: what every number is built from

A truck's visit is split into six steps

Each time a truck moves to the next step, the app saves the time. The minutes between two saved times are the minutes for that step.

Step	Starts when…	Ends when…	Target
1. Gate check-in	The truck arrives at the gate	The guard finishes checking it in	3–5 min
2. Dock assignment	Check-in is done	Shipping gives the truck a dock door	15–30 min
3. Pre-loading	The dock door is assigned	Loading or unloading starts	5–10 min
4. Loading / unloading	Loading starts	Loading finishes	30–60 min
5. Inspection	Loading finishes	The loading inspection is saved	10–15 min
6. Gate checkout	The inspection is saved	The truck is checked out at the gate	3–5 min

Two combined times

Dock handling time = Pre-loading + Loading / unloading + Inspection (steps 3, 4 and 5: the time spent at the dock door).
Gate-to-gate time = all six steps added together (from arriving at the gate to leaving it).

Example: a truck takes 4 min at check-in, 20 min waiting for a dock, 8 min pre-loading, 45 min loading, 12 min inspection and 4 min at checkout. Its dock handling time is 8 + 45 + 12 = **65 min**. Its gate-to-gate time is 4 + 20 + 65 + 4 = **93 min**.

Which trucks are counted

- **Every truck checked in** is used for the counting numbers: *Trucks handled*, *On-time arrivals*, trucks per day and arrivals by hour.
- **Only "fully timed" trucks** are used for the time and money numbers: *Gate-to-gate*, *Dock handling*, *Within target*, *Detention* and the step charts. A truck is fully timed when all seven times above were saved and it has checked out. Drop trailers never are, because they don't go through the dock steps.
- The top of the page shows how many trucks were fully timed, for example "42 of 50 trucks fully timed".

Date range and sample data

- The buttons **Last 7 / 14 / 30 days** pick which days are included, always ending today.
- If there are no fully timed trucks in that range yet, the page shows built-in **sample data** instead and labels it "Sample data" in yellow, so it is never mistaken for real numbers.
- **Averages** mean: add up the value for every truck, then divide by the number of trucks.

Arriving early, on time or late

Minutes vs appointment = arrival time − appointment time.
More than 15 min early = **Early** · within 15 min either side = **On time** · more than 15 min late = **Late**.

A truck booked for 09:00 that arrives at 09:10 is +10 min, so it is on time. One that arrives at 09:20 is +20 min, so it is late. Older visits with no appointment time count as on time.

2. The six KPI cards

Each card shows one number for the whole date range, a small line (sparkline) for the last 7 days, and a green or red label. Clicking a card opens its detail page with more charts and the list of trucks behind the number.

Gate-to-gate turnaround

How long a truck stays with us, on average.

Calculation: add up the gate-to-gate minutes of every fully timed truck, then divide by the number of those trucks. Rounded to whole minutes.

Label: "On target" if the average is 130 min or less, "Over target" if it is more. Target range is 65–130 min.

Example: three trucks take 90, 110 and 160 min. Average = (90 + 110 + 160) ÷ 3 = **120 min**, so On target.

Dock handling time

How long a truck spends at the dock door, on average.

Calculation: add up the dock handling minutes (pre-loading + loading + inspection) of every fully timed truck, then divide by the number of trucks.

Label: "On target" at 85 min or less, "Over target" above. Target range is 45–85 min.

On-time arrivals

The share of trucks that arrived within 15 minutes of their booked time.

Calculation: number of trucks marked On time ÷ number of trucks checked in, shown as a percentage.

Example: 50 trucks checked in, 41 were on time. 41 ÷ 50 = **82%**. Early and late trucks both count against this number.

Trucks handled

How busy we were.

Calculation: a simple count of every truck checked in at the gate in the date range: deliveries, pickups and drop trailers.

If a visit has several purposes ticked, it is counted under the first one.

Within turnaround target

The share of trucks that were done in time.

Calculation: number of fully timed trucks with a gate-to-gate time of 130 min or less ÷ number of fully timed trucks, as a percentage.

Label: "On target" at 90% or more, "Below 90%" otherwise.

Example: 40 fully timed trucks, 36 finished in 130 min or less. 36 ÷ 40 = **90%**, so On target.

Detention risk

An estimate of the money carriers could charge us for holding their trucks too long.

For each truck:
1. Start the clock at the **later** of the appointment time and the arrival time. (A truck that comes early doesn't start the clock before its booking.)
2. Stop the clock when the truck checks out.
3. The first **2 hours are free**. Every hour after that costs **$75** (charged by the minute, rounded to whole dollars).
4. A truck that arrived **more than 15 min late is charged $0**, because the delay was the carrier's.

The card adds up the estimate for every fully timed truck in the range.

Label: "Money at risk" if the total is above $0, "No detention" if it is $0.

Example: booked 09:00, arrived 08:55, left at 11:45. The clock starts at 09:00 (the later time) and runs 2 h 45 min. Minus 2 h free = 45 min over. 45 ÷ 60 × $75 = **$56**.
The same truck arriving at 09:20 (20 min late) would be **$0**.

The summary sentence at the top

The sentence under the page title is built from the same numbers: minutes saved and detention avoided since going digital (see section 3), how many trucks went over 130 min, the step that is furthest over its target, and the day with the highest average gate-to-gate time.

"Furthest over its target" means the step with the biggest average ÷ top-of-target. Inspection averaging 18 min against a 15 min top (1.2×) beats Loading averaging 66 min against 60 (1.1×).

3. Impact of going digital

This section compares two weeks:

- **Manual week:** a week from the old paper log, imported with "Import manual week". Until one is imported, a sample manual week stands in and is labeled.
- **Digital week:** the latest 7 days of fully timed trucks in this app, counting back from the most recent day with data.

Saved = manual week value − digital week value. **Change %** = (digital − manual) ÷ manual.

Faster per truck

Calculation: average gate-to-gate minutes in the manual week − average in the digital week.

Example: 150 min → 110 min means **40 min saved** per truck, a change of (110 − 150) ÷ 150 = −27%.

Detention avoided this week

Calculation: total detention estimate for the manual week − total for the digital week. The "a year at this pace" line multiplies each week by 52.

Example: $900 → $150 means **$750 avoided**, about $750 × 52 = $39,000 a year.

Shorter wait for a dock

Calculation: average Dock assignment minutes (step 2) in the manual week − average in the digital week.

Trucks within target

Calculation: the share of trucks at 130 min or less gate to gate, worked out separately for each week. The tile shows the digital week's share and "Up from" the manual week's share.

Phone calls no longer needed

Calculation: the total phone calls between gate and shipping recorded in the manual week's log. The digital week counts as zero, because updates now appear on the shared screen. The same is done for paper handoffs and for minutes spent searching the yard for a truck (shown in hours).

On the detail page, staff time = calls × minutes per call + handoffs × minutes per handoff + yard-search time. Minutes per call (default 3) and per handoff (default 2) are estimates you can change on that page.

Truck hours freed up

Calculation: add up the gate-to-gate minutes of every truck in each week and divide by 60 to get hours on site. Saved = manual week hours − digital week hours.

This is a total, not an average, so a week with more trucks naturally has more hours.

Charts in this section

Chart	What each value is
Every stage, manual vs digital	For each of the six steps: the average minutes per truck in the manual week and in the digital week.
Gate to gate by weekday	Average gate-to-gate minutes for trucks on each weekday, manual vs digital. Only weekdays that have trucks in both weeks are shown. The red dashed line is the 130 min maximum.
Detention by weekday	Total detention estimate for each weekday, manual vs digital.

4. Where the time goes

Chart	What each value is
Average minutes per stage, by day	One bar per day. Each coloured piece is the average minutes for one step across that day's fully timed trucks; the full bar is that day's average truck. The red dashed line is the 130 min maximum.
Each stage against its target	For each step: the average minutes over the whole date range (the bar), drawn against its target range (green band). Red means the average is above the top of the target. Hovering shows "Within": the share of trucks at or under the top of the target for that step.

5. Arrivals and volume

Chart	What each value is
Trucks per day by visit type	A count of trucks checked in each day, split into deliveries, pickups and drop trailers.
Arrival vs appointment	The number of trucks that were early, on time and late (see section 1). The middle shows the on-time percentage, the same as the On-time arrivals card.
Arrivals by appointment hour	Trucks counted by the hour of their appointment (09:00 = booked 09:00–09:59). A bar is red when that hour's fully timed trucks averaged more than 130 min gate to gate.
Does arriving late slow the whole visit?	One dot per fully timed truck: across is minutes vs appointment (left = early, right = late), up is gate-to-gate minutes. The green band is the on-time window (±15 min). Red dots are trucks over 130 min.

6. All targets in one place

What	Target / rule
Gate-to-gate turnaround	65–130 min (over 130 = over target)
Dock handling time	45–85 min (over 85 = over target)
Within turnaround target	90% or more of trucks at 130 min or less
On time	Arrival within ±15 min of the appointment
Detention	2 h free from the later of appointment and arrival, then $75 an hour; trucks more than 15 min late aren't charged
Steps	Check-in 3–5 · Dock assignment 15–30 · Pre-loading 5–10 · Loading 30–60 · Inspection 10–15 · Checkout 3–5 (min)

Detention figures are estimates using the rule above, not actual carrier invoices.